# Mikhail Maslov, PhD

Quantitative Research | Systematic Strategies | Portfolio & Risk Modelling

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Theoretical physicist with experience in mathematical modelling, scientific computing, and machine learning. Seeking quant research roles in systematic strategies, portfolio and risk modelling. Willing to relocate Europe-wide.

## EDUCATION

**PhD in Theoretical Physics**, Institute of Science and Technology Austria (2018–2024)
Thesis: *"Emergent physics of rotating quantum impurities in many-body environments"*
Research focus: group theory, mathematical physics, numerical modelling, and PINNs.

**BSc in Mathematics and Physics**, Moscow Institute of Physics and Technology (2014–2018)
Graduated with Honors (GPA 8.6/10), received Abramov and State Academic fellowships.

## EXPERIENCE

**Researcher** at Institute of Science and Technology Austria (2018–present)

- built neural-network solvers for differential equations in quantum few-body systems using reinforcement learning and Monte Carlo sampling
- developed a universal group-theoretical framework for analyzing angular momentum exchange in optical systems, applicable to many research fields in physics
- developed a symmetry-aware mathematical model for optical sensing of material chirality
- formulated analytical models for controlling quantum many-body systems; assessed their feasibility numerically using variational methods
- 7 publications, 140 citations; reviewer for *Phys. Rev. A* and *Phys. Rev. Research*

**Undergraduate Researcher** at Moscow Institute of Physics and Technology (2016–2018)

- developed a theoretical framework based on coupled partial differential equations to analyze plasmonic instabilities in nanostructured semiconductor systems
- designed and maintained a MATLAB codebase for electrodynamics simulations of two-dimensional electron systems

**Intern** at Helmholtz Zentrum Dresden-Rossendorf (summer 2017)

- designed and optimized code for molecular dynamics simulations on an HPC cluster

**Intern** at Deutsches Elektronen-Synchrotron, Hamburg (summer 2016)

- developed MATLAB code for simulating crystallographic data using Monte Carlo method and comparing it to the experimental data from free-electron laser via Fourier analysis

**Student Representative** at Institute of Science and Technology Austria (2019–2022)

- designed curriculum and taught atomic and condensed matter physics to ~50 PhD students
- managed curriculum and student infrastructure development (~500 PhD students)
- organized networking and scientific outreach events

**Teaching assistant** at Moscow Institute of Physics and Technology (2014–2019)

- taught mathematics, physics and computer science to ~100 high-school students

## SKILLS

**Mathematics & modelling:** linear algebra, tensor calculus, probability theory, differential equations, mathematical physics, group theory, differential geometry, quantum algorithms

**Numerical methods & ML:** Monte Carlo methods, variational methods, finite-difference methods, Fourier analysis, optimization, supervised and reinforcement learning

**Programming:** Python (NumPy, SciPy, PyTorch), MATLAB, Wolfram Mathematica, bash

**Computing:** vectorization, parallel computing, HPC, Linux, containerization, automation

**Selected interests:** game theory, decision theory, auction theory; market making and risk management; macroeconomics and public policy